BenHalluEval v2.0: Corrected Master Architecture Specification for Bengali LLM Hallucination Detection and Mitigation

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Abstract

This document provides the corrected mathematical formulation and design specification for the BenHalluEval v2.0 6-Stage Master Architecture. It audit-corrects mathematical notation, task boundaries, score definitions, routing logic, logit-level context-contrastive decoding (CAD), split-conformal prediction sets, operational failure policies, and dual-track BenHalluScore (BHS) metrics across low-resource Bengali LLM hallucination evaluation and active mitigation.

1 System Task Boundary & Thesis Label Conventions

Let $x_i = (q_i, r_i, c_i)$ be an input query triple, where q_i is the prompt, r_i is the candidate response, and $c_i \in \mathcal{C} \cup \{\varnothing\}$ is optional grounding context. The detector estimates $\hat{y}_i = f_\theta(x_i) \in \{0, 1\}$. The thesis label convention is fixed throughout:

$$y_i = 0 \iff \text{faithful / non-hallucinated}, \qquad y_i = 1 \iff \text{hallucinated}. \tag{1}$$

Detection and mitigation are distinct tasks:

- **Detection:** Classify the original candidate response r_i.
- **Mitigation:** If $\hat{y}_i = 1$, optionally generate a revised answer r_i^*. A repaired answer r_i^* must be evaluated as a newly generated output; it cannot inherit gold labels or self-certify.

2 Corrected BenHalluScore (BHS) Formulation

Let $A = \{i : y_i = 0\}$ ($N_A = |A| > 0$) and $B = \{i : y_i = 1\}$ ($N_B = |B| > 0$). Track-A False Positive Rate (False Alarms) and Track-B False Negative Rate (Missed Hallucinations):

$$\mathrm{FPR}_A = \frac{1}{N_A} \sum_{i \in A} \mathbf{1}[\hat{y}_i = 1], \qquad \mathrm{FNR}_B = \frac{1}{N_B} \sum_{i \in B} \mathbf{1}[\hat{y}_i = 0]. \tag{2}$$

The weighted and default balanced BenHalluScore (lower is better):

$$\mathrm{BHS}_{w_A, w_B} = 100 \left(w_A \, \mathrm{FPR}_A + w_B \, \mathrm{FNR}_B \right), \qquad \mathrm{BHS} = \frac{100}{2} \left(\frac{W_A}{N_A} + \frac{W_B}{N_B} \right). \tag{3}$$

2.1 Operational Failure-Aware BHS

For system outcome $\tilde{y}_i \in \{0, 1, \perp\}$ where $\perp$ denotes inference/parsing failure:

$$e_i^{(A)} = \mathbf{1}[\tilde{y}_i \in \{1, \perp\}], \; i \in A; \qquad e_i^{(B)} = \mathbf{1}[\tilde{y}_i \in \{0, \perp\}], \; i \in B. \tag{4}$$

$$\mathrm{BHS}_{\mathrm{op}} = 100 \left(w_A \frac{\sum_{i \in A} e_i^{(A)}}{N_A} + w_B \frac{\sum_{i \in B} e_i^{(B)}}{N_B} \right). \tag{5}$$

If either track is absent, aggregate BHS is **undefined**.

3 Stage 1: Raw-Text Dual-View Normalization & Audit

Raw text S_i^{raw} is preserved. Derived views:

$$S_i^{\text{native}} = N_{\text{bn}}(S_i^{\text{raw}}), \qquad S_i^{\text{code}} = N_{\text{code}}(S_i^{\text{raw}}). \tag{6}$$

Transducer precedence:

$$\Phi(w) = \begin{cases} w, & w \in \mathcal{V}_{\text{English}} \cup \mathcal{V}_{\text{technical}} \\ \text{Translit}_{\text{BN}}(w), & w \in \mathcal{V}_{\text{RomanBangla}} \\ w, & \text{otherwise.} \end{cases} \tag{7}$$

Token Fertility Ratio ($W(S) > 0$):

$$F_\tau(S) = \frac{|\operatorname{Tok}_\tau(S)|}{W(S)}. \tag{8}$$

13-Token Overlap Audit ($I_{13}(x_i)$):

$$I_{13}(x_i) = \mathbf{1}\left[H_{13}(N(q_i \parallel r_i \parallel c_i)) \cap \mathcal{G}_{\text{train}} \neq \varnothing\right]. \tag{9}$$

4 Stage 2: Regime-Aware Contrastive RAG

Evidence pool:

$$\mathcal{D}_i = \begin{cases} \text{Segment}(c_i), & c_i \neq \varnothing \\ \mathcal{D}_{\text{external}}^{\text{approved}}, & c_i = \varnothing. \end{cases} \tag{10}$$

Atomic claims $\text{Claims}(r_i) = \{c_{i1}, \ldots, c_{iK_i}\}$. Contrastive query pairs: $q_{ik}^+ = Q(q_i, c_{ik})$ and $q_{ik}^- = Q(q_i, \neg c_{ik})$. Polarity-preserving RRF ($s \in \{+, -\}$):

$$\text{RRF}^s(d \mid c_{ik}) = \sum_{m \in \mathcal{M}} \frac{\mathbf{1}[d \in L_{m,q_{ik}^s}]}{60 + \text{rank}_{m,q_{ik}^s}(d)}. \tag{11}$$

Documents retrieved by q_{ik}^- are potential counterevidence, not positive support.

5 Stage 3: Per-Claim NLI & Conflict-Aware Routing

For atomic scores $p_{ik}^{\text{ent}} = \max_j P_{\text{NLI}}(\text{entail} \mid e_{ikj}, c_{ik})$ and $p_{ik}^{\text{con}} = \max_j P_{\text{NLI}}(\text{contradict} \mid e_{ikj}, c_{ik})$:

$$\text{SMin}_\tau^{\text{ent}}(r_i) = -\tau \log \sum_{k=1}^{K_i} w_{ik} \exp\left(-\frac{p_{ik}^{\text{ent}}}{\tau}\right), \qquad \text{SMax}_\tau^{\text{con}}(r_i) = \tau \log \sum_{k=1}^{K_i} w_{ik} \exp\left(\frac{p_{ik}^{\text{con}}}{\tau}\right). \tag{12}$$

Retain hard extrema $E_{\min} = \min_k p_{ik}^{\text{ent}}$ and $C_{\max} = \max_k p_{ik}^{\text{con}}$. Tri-State Conflict-Aware Routing Rule:

$$R_i = \begin{cases} \text{fast-faithful}, & c_i \neq \varnothing \wedge \neg\,\text{isMath}(x_i) \wedge S_{\text{ent}} \geq \gamma_E \wedge C_{\max} \leq \lambda_C \\ \text{fast-hallucinated}, & c_i \neq \varnothing \wedge \neg\,\text{isMath}(x_i) \wedge S_{\text{con}} \geq \gamma_C \wedge E_{\min} < \gamma_E \\ \text{full-verification}, & \text{otherwise.} \end{cases} \tag{13}$$

6 Stage 4: Two-Label Score & Symbolic Sandbox

Two-label normalized first-token score ($m^{(2)}(S) = |2s_h^{(2)}(S) - 1|$):

$$s_h^{(2)}(S) = \frac{\exp L_h(S)}{\exp L_h(S) + \exp L_f(S)} = \sigma\left(L_h(S) - L_f(S)\right). \tag{14}$$

Fixed code-mixed score: $p_{\text{cm}} = \sigma(L_{h,\text{cm}} - L_{f,\text{cm}})$. Cross-view Jensen-Shannon Divergence (D_{script}):

$$D_{\text{script}}(x_i) = \frac{1}{2} D_{\text{KL}}(P_i \parallel M_i) + \frac{1}{2} D_{\text{KL}}(Q_i \parallel M_i). \tag{15}$$

Symbolic execution result $z_i^{\text{exec}} \in \{\text{verified}, \text{contradicted}, \text{unresolved}, \text{timeout}, \text{exception}\}$. Timeouts and exceptions must not default to faithful.

7 Stage 5: Subspace-Compressed Meta-Classifier & Conformal Prediction

Out-of-fold lexical risk score: $z_i^{\text{lex,OOF}} = P_{g(i)}(Y = 1 \mid v_i^{\text{lex}})$. Dense feature vector $v_i \in \mathbb{R}^{31}$. Joint BHS threshold selection:

$$(\theta_o^*, \theta_c^*) \in \arg\min_{(\theta_o,\theta_c)\in\Theta^2} \widehat{\text{BHS}}_{\text{dev}}(\theta_o, \theta_c). \tag{16}$$

Valid Split-Conformal Prediction Set ($a_i = 1 - \hat{p}_{y_i}(x_i)$):

$$\Gamma_\alpha(x) = \{y \in \{0, 1\} : 1 - \hat{p}_y(x) \le \hat{q}_{1-\alpha}\}. \tag{17}$$

8 Stage 6: Logit-Level CAD & Operational Mitigation

Contradicted claims: $\mathcal{C}_i^{\text{bad}} = \{c_{ik} : p_{ik}^{\text{con}} > \tau_{\text{con}}\}$. Logit-Level Context-Contrastive Decoding (CAD):

$$\ell_{\text{CAD}}(y_t) = (1 + \alpha)\ell(y_t \mid q_i, c_i, r^*_{i,<t}) - \alpha\ell(y_t \mid q_i, r^*_{i,<t}), \qquad P_{\text{CAD}}(y_t) = \text{softmax}\left(\ell_{\text{CAD}}/T\right)_{y_t}. \tag{18}$$